AI Opportunity Report

Cynbit Studio – AI Resume & Job Matching Intelligence · Prepared for the Cynbit Technologies AI/ML Engineer Intern assessment · 30 October 2026

1. Executive summary

Students and early-career candidates apply widely but rarely know **which roles they genuinely fit**, **which skills are missing**, and **what to learn next**. Cynbit Studio is a working prototype that turns a resume and a set of job descriptions into an explainable match score, a ranked job list, a prioritised skills gap, a week-by-week learning plan, resume fixes and a career-readiness index. It runs locally with open models (Sentence Transformers *all-MiniLM-L6-v2*, spaCy), so it is cheap to operate and privacy-friendly, with optional Gemini coaching on top.

For Cynbit Technologies, the prototype is a reusable **AI module** that can be offered to colleges, placement cells, training institutes and recruiting clients, embedded into client websites, and used to generate useful, search-friendly career content.

2. The problem

The India Skills Report 2025 estimates graduate employability at about **54.8%**, meaning that roughly 45% of assessed graduates are not yet considered job-ready. The gap is less about effort than about direction: candidates lack clear, personalised feedback.

- **Opaque screening:** keyword-based ATS filters reject resumes without telling candidates why.
- **Unclear targeting:** students apply to roles that don't match their profile and miss ones that do.
- **Generic learning:** course recommendations aren't tied to the skills specific jobs require.
- **Weak resumes:** duty-style bullets with no measurable impact hide real ability.
- **Limited placement-cell capacity:** one officer may serve hundreds of students.

Source: India Skills Report 2025 (Wheebox / CII / AICTE), as reported by Deccan Herald and Swarajya, Nov 2024.

3. Target users and value

Segment	Pain point	Value delivered
Students & freshers	Don't know fit or gaps	Ranked jobs, missing skills, learning plan, resume fixes
Colleges / placement cells	Too many students, little time	Batch readiness dashboards, cohort skill-gap analytics
Training institutes	Generic curricula	Data on which skills are demanded vs. missing
Recruiters / SMEs	Noisy applicant pools	Explainable shortlisting aid (human-in-the-loop)
Cynbit's web clients	Static career pages	Embeddable 'check your fit' widget on job pages

4. Solution delivered in this prototype

Capability	How it works (real NLP, no hard-coded scores)
Resume parsing	PyMuPDF (reading-order text) and python-docx (incl. tables); unicode/bullet clean-up
Skill & experience extraction	90-skill taxonomy (~430 surface forms) with special-character-safe matching; date-range parsing with overlap merging; degree & CGPA detection; spaCy NER
Job description analysis	Required vs. preferred skills by section and cue phrases; min. years, degree, seniority; spaCy noun-chunk key phrases

Capability	How it works (real NLP, no hard-coded scores)
Semantic matching	MiniLM embeddings: whole-document cosine + per-requirement best-evidence alignment
Explainable score	40% semantic · 35% skills · 15% experience · 10% education, each shown with reasons
Gap & learning plan	Missing / related skills; priority × effort scheduling; curated free resources
Multi-job ranking	Ranking table, grouped chart, skill-demand heatmap, CSV export
Readiness dashboard	Composite index, radar, category coverage, downloadable PDF report
Resume improvement	Impact, action verbs, weak phrases, passive voice (spaCy), structure, keywords

5. Why this approach

- **Explainable:** every score decomposes into components and evidence lines a student can act on.
- **Low cost:** MiniLM (~23M parameters) runs on CPU in milliseconds per sentence; no per-request LLM fees.
- **Private by design:** resumes stay local; history stores only a fingerprint and scores; PII is redacted before any optional LLM call.
- **Robust:** automatic offline fallback (TF-IDF + LSA) keeps the product usable without model downloads.

6. Integration with Cynbit Studio

Cynbit Technologies' public profile lists SEO, web design, UX/UI and custom software development. The plan below assumes Cynbit Studio is (or becomes) the company's product/delivery platform for client web solutions; it requires **no change to Cynbit's current website** and can be adopted incrementally.

Phase	Deliverable	Technical approach	Effort
1 · Service	Matching API	FastAPI wrapper around cynbit.analyze(); Docker; /parse /match /rank /recommend	2–3 wks
2 · Widget	Embeddable 'Check your fit'	JS widget calling the API; JWT/SSO; brandable per client	2–4 wks
3 · Scale	Job index	PostgreSQL + pgvector, job embeddings computed once, nightly feeds	3–4 wks
4 · Analytics	Placement-cell dashboard	Cohort readiness, skill-demand trends, exports	3–4 wks
5 · Learning loop	Calibrated scoring	Fit component weights to recruiter shortlist outcomes	ongoing
6 · GenAI	Coaching & content	Gemini/LLM bullet rewrites, cover letters, interview prep, SEO-friendly skill-gap articles	2–3 wks

Business models: (a) B2B SaaS licence for colleges and training institutes (per-student/year); (b) white-label module bundled into Cynbit web projects for HR, staffing and education clients; (c) lead generation and SEO value from public, privacy-safe skill-gap insights; (d) recruiter add-on for explainable pre-screening.

7. Success metrics

Metric	Target for a pilot (one college, one semester)
Activation	≥ 60% of final-year students upload a resume

Metric	Target for a pilot (one college, one semester)
Engagement	≥ 2 job comparisons per active student; ≥ 30% revisit within 30 days
Outcome	Readiness index +10 points median after 8 weeks; ↑ interview-call rate vs. previous batch
Quality	Top-1 job ranking agrees with placement officer judgement in ≥ 80% of sampled cases
Trust	Zero raw resumes stored server-side; documented consent under India's DPDP Act, 2023

8. Risks and responsible AI

Risk	Mitigation
Bias against non-standard resumes or regional phrasing	Skill-alias taxonomy, semantic matching, human review; audit scores by college/region
Over-reliance on a single number	Component scores + evidence; UI states scores are guidance, not decisions
Keyword gaming	Requirement-level semantic evidence and 'back up listed skills' checks
Privacy / data protection	Local processing, fingerprint-only history, PII redaction, consent, retention limits
LLM hallucination (optional coaching)	LLM never sets scores; prompts forbid inventing facts and use [X] placeholders

9. Next steps

- Validate on 100–200 real, consented resume–job pairs labelled by placement officers; calibrate weights.
- Ship the FastAPI service and widget (Phases 1–2) for a single pilot institution.
- Extend the taxonomy with Cynbit's client domains and regional job titles; add OCR for scanned resumes.